Confusion Matrix –Random Forest

A **confusion matrix** is a table that shows how a classification model is performing by comparing **actual vs predicted** values

```
[[78  7]
 [ 6 43]]
TP= 78
FP=7
TN = 43
FN=6
Total =134
```

Meaning of Each Term

- TP (True Positive): Model correctly predicts class 1
- TN (True Negative): Model correctly predicts class 0
- FP (False Positive): Model predicts 1 but actually 0 (False Alarm)
- FN (False Negative): Model predicts 0 but actually 1 (Missed Case)

	precision	recall	f1-score	support
Class 0 → 0	0.93	0.92	0.92	85
Class 1 → 1	0.86	0.88	0.87	49
accuracy			0.90	134
macro avg	0.89	0.90	0.90	134
weighted avg	0.90	0.90	0.90	134

Metrics Derived from Confusion Matrix

- Accuracy → overall correctness
- Precision → correctness of positive predictions
- Recall → ability to find positives
- Specificity → ability to find negatives

1. How is Precision calculated for Class 1?

Solution:

Formula:

Precision=TP/(TP+FP)

TP = 43

FP=7

43/50=0.86

2. How is Recall calculated for Class 1?

Solution:

Formula:

Recall=TP/(TP+FN)

TP=43

FN=6

Recall = 0.88

3. How is F1-measure calculated?

Solution:

Formula:

F1=2·(Precision*recall)/(Recall+Precision)

=2*0.86*0.88/(0.86+0.88)

=1.51/1.74

=0.87

4. Can you derive Accuracy from this report?

Solution:

Formula:

Accuracy=(TP+TN)/Total

TP+TN=78+43=121

Total=134

Accuracy= 0.90

5. What is Specificity?

Solution:

Formula:

Specificity=TN/(TN+FP)

=43 / (43 + 7) ≈ 0.86

6. Can you reconstruct the Confusion Matrix (approx)?

Solution:

From support:

- Class 0 → 85
- Class 1 → 49

From recall:

- Class 1 recall = 0.88 → TP ≈ 43, FN ≈ 6
- Class 0 recall = 0.92 → TN ≈ 78, FP ≈ 7

	Pred 0	**Pred 1**
Actual 0	76	7
Actual 1	6	43

7. How is Macro Average calculated-precision?

Solution:

Formula:

Macro Average= (P(1)+P(0))/2

=(0.93+0.86)/2

Macro Average = 0.895

8. How is Weighted Average-Precision calculated?

Solution:

Formula:

WA = P(0) *Support (0) + P(1)* Support(1)/ Total

= 0.93 × 85 + 0.86 × 49) / 134 ≈ 0.904

9. How do precision and recall relate mathematically?

Solution:

- Precision depends on **false positives**

- Recall depends on **false negatives**

10. How do you improve model using confusion matrix?

Solution:

- Analyze FP & FN
- Adjust threshold
- Use class weights
- Collect more data